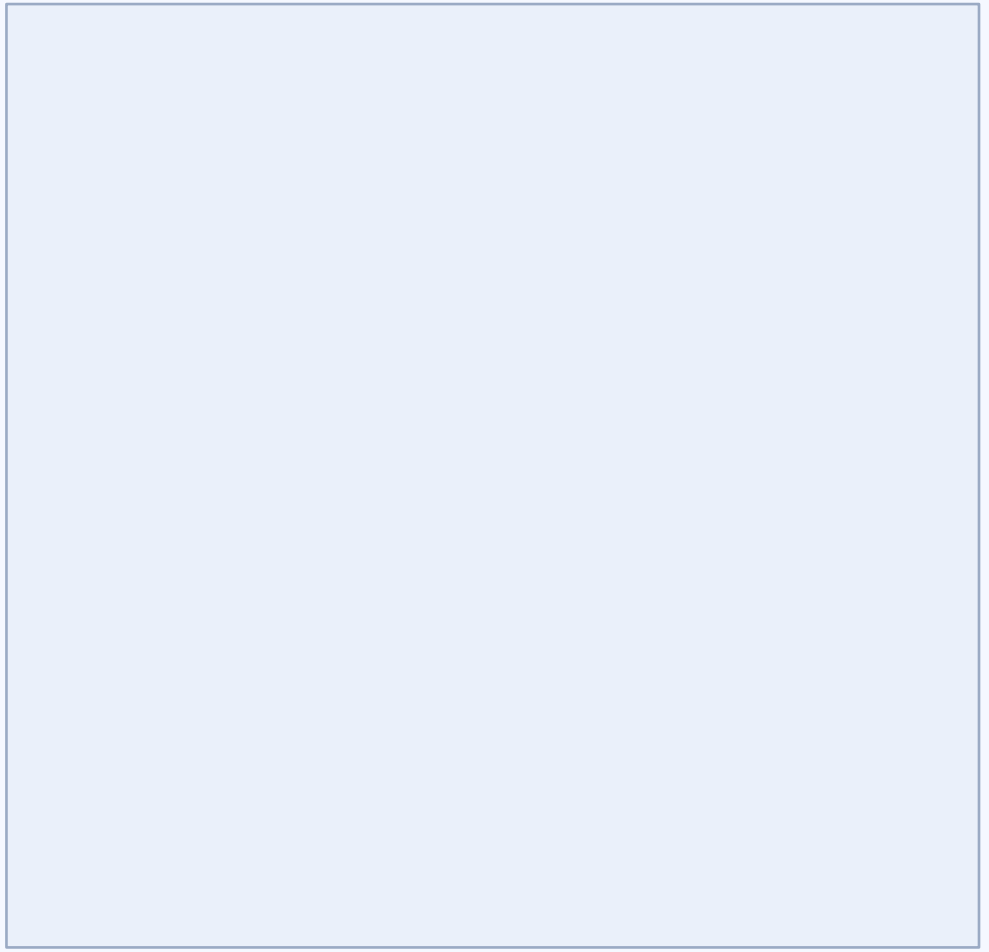
DC LAB / ELECTRONICS + CONTROL

[Crash course title goes here]

[A practical question this session answers.]

[Session 01] / [date] / [duration]



Today, we will [build / measure / debug]

01	**Understand**	[The one idea learners need]
02	**Trace**	[Follow the signal or data path]
03	**Build**	[The thing learners will make]
04	**Verify**	[A measurement that proves it works]

>_

[What changes when we remove delay()?]

[Ask for a prediction before showing the answer.]

[Name the thing. Explain what it does.]

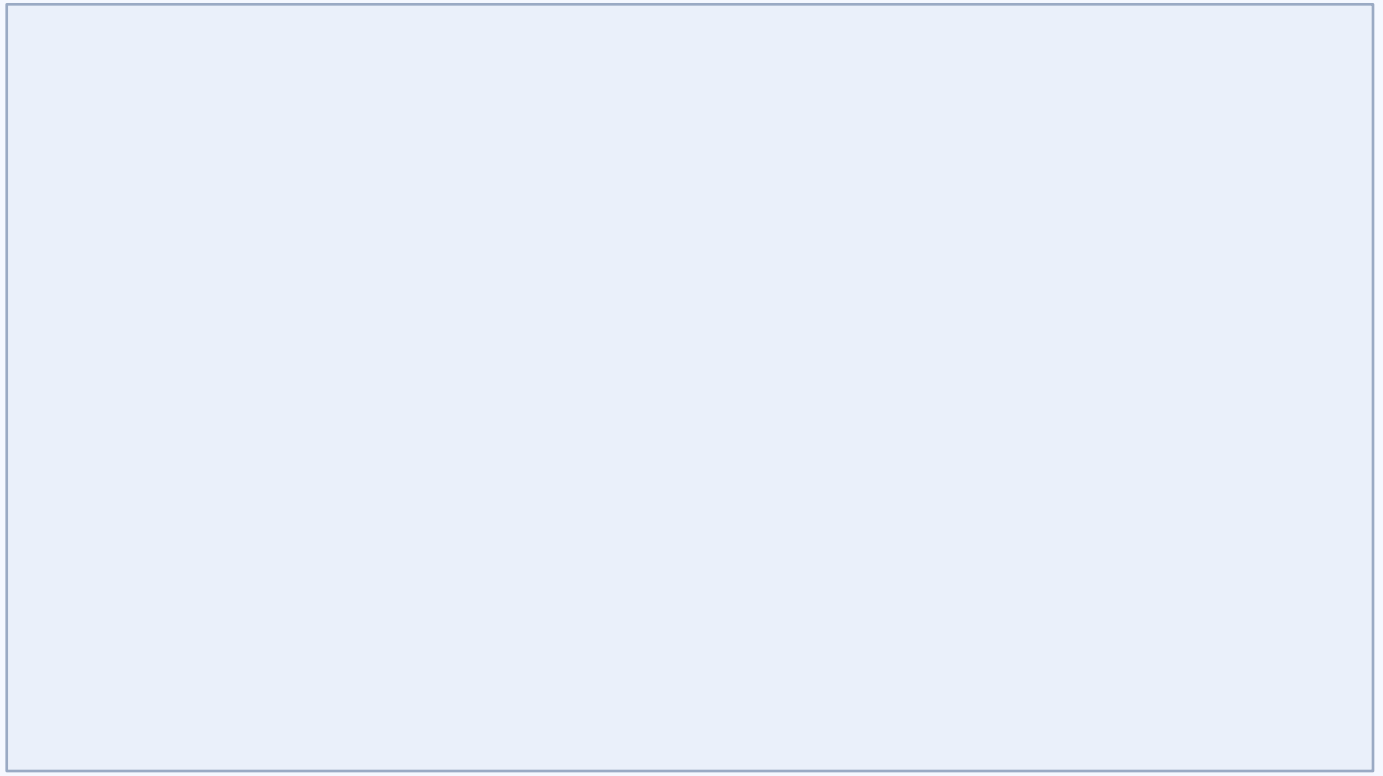
[One-sentence mental model]

[What goes in?]

[What changes?]

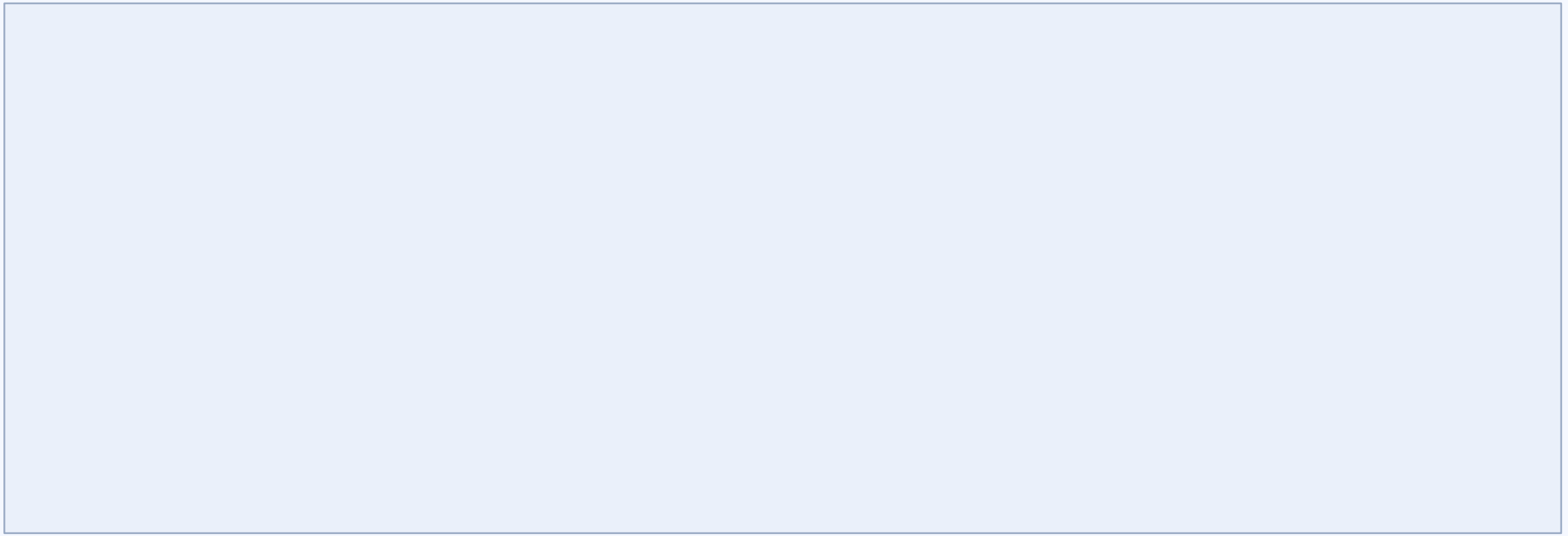
[What comes out?]

[One important constraint]



[Figure / source / what to notice]

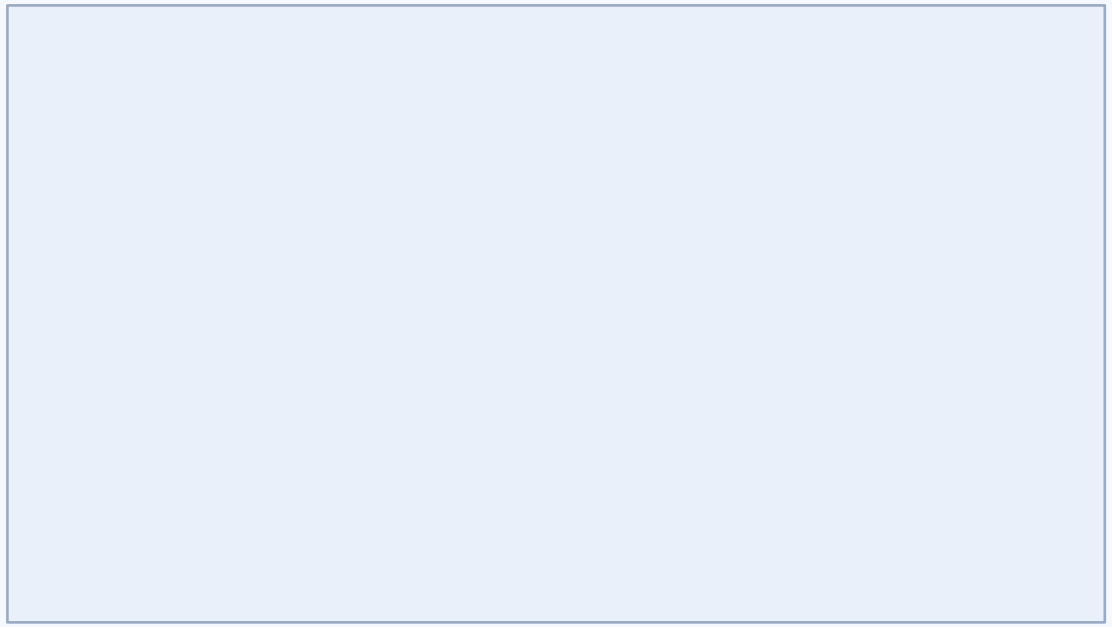
[Give the viewer a reason to look]



[Figure 01] [What this figure demonstrates] / [source]

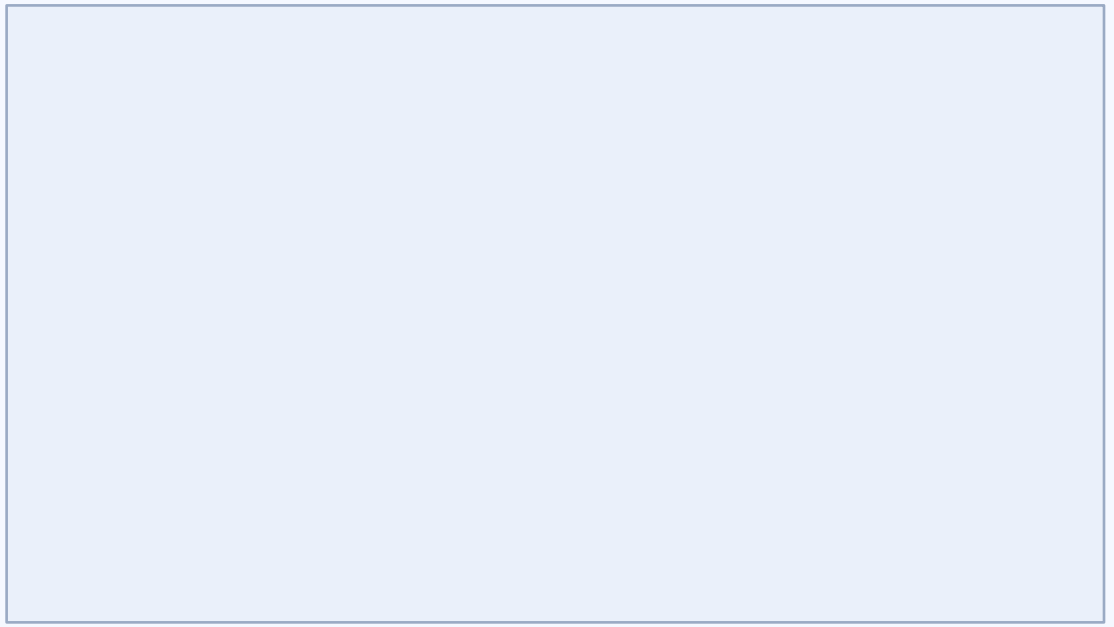
[What changed between these two?]

[A / before]



[What to notice]

[B / after]



[What to notice]

[One small idea in code]

C++ / timing.cpp

```cpp
// One quick task; then return to the loop.
constexpr uint32_t PERIOD_MS = 10;
uint32_t previous = 0;

void loop() {
  const uint32_t now = millis();
  if (uint32_t(now - previous) >= PERIOD_MS) {
    previous = now;
    read_sensor();
  }
  poll_serial();
}
```

[Read it in this order]

01 [Read the time]

02 [Check elapsed time]

03 [Do a short task]

[A full-width code walkthrough]

```
// Keep each task short enough to return promptly.
void loop() {
  const uint32_t now = millis();
  update_inputs(now);
  update_control(now);
  update_outputs(now);
  service_telemetry(now);
}
```

[Ask: what runs next, and why?]

[Trace the path from input to output]

[Label every arrow with what actually crosses the boundary.]



[Takeaway: where is the interface most likely to fail?]

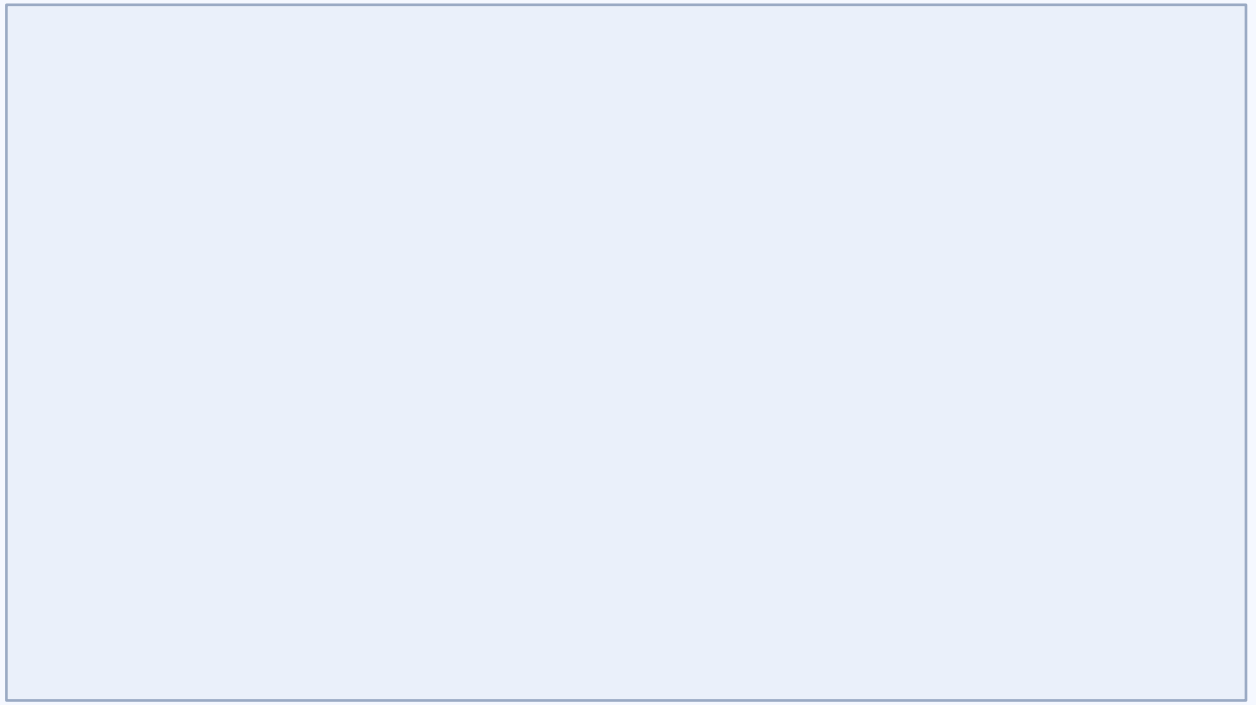
[What does the equation predict?]

$$D = t_{on} / T$$

D [duty ratio]

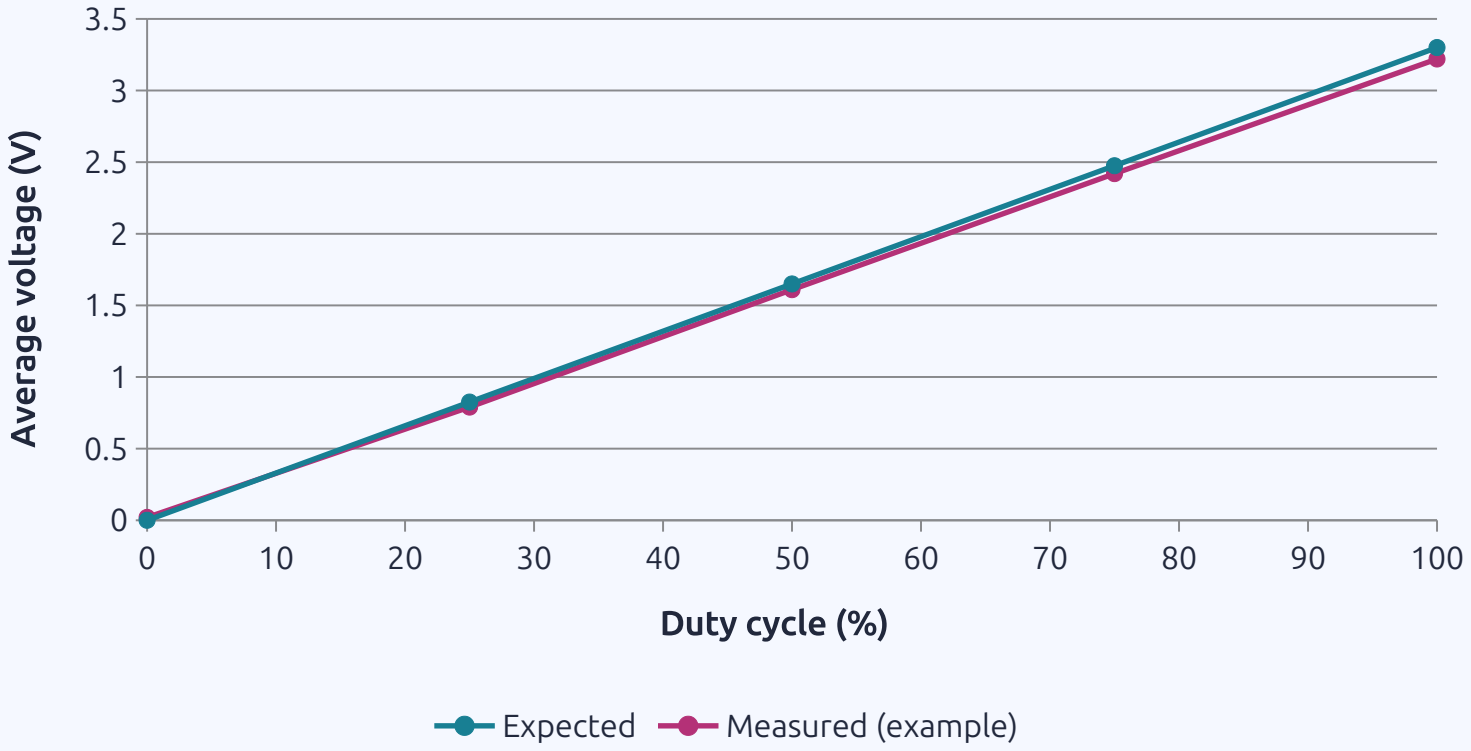
t_on [high time]

T [period]



[State assumptions and units.]

[Does the measurement match the prediction?]



[Read the trend]

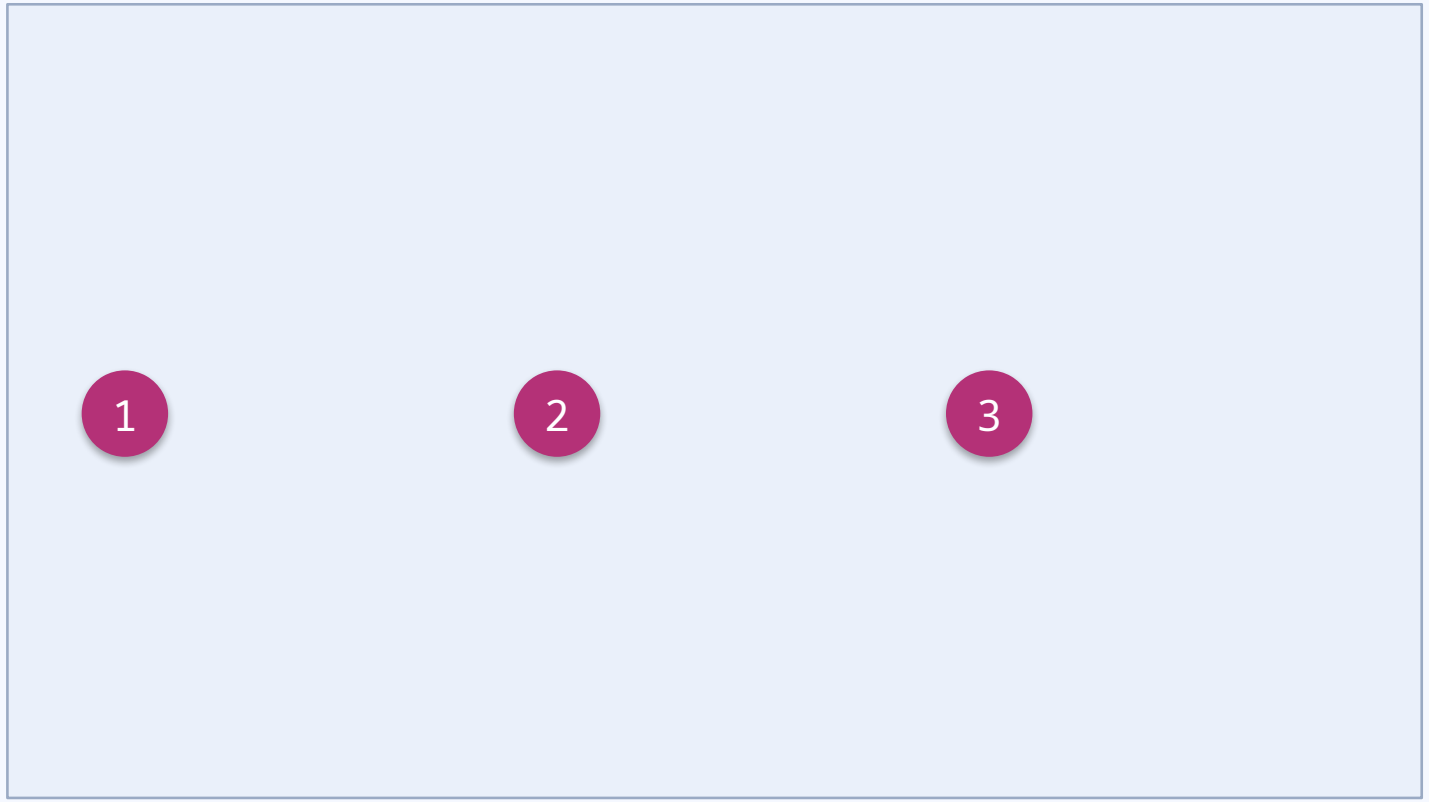
[Expected behaviour]

[Observed deviation]

[Likely cause]

ILLUSTRATIVE DATA / replace with your own measurements

[Orient the board before connecting it]



01 **[Connection 1]**

[Pin / voltage / direction]

02 **[Connection 2]**

[Pin / voltage / direction]

03 **[Connection 3]**

[Pin / voltage / direction]

[Build a small thing. Prove it works.]

01 [Connect / configure]

02 [Change one variable]

03 [Measure and record]

DONE WHEN

[A visible result or measurable pass condition]

[Timebox: 10 min]

[Start with evidence, then change one thing]

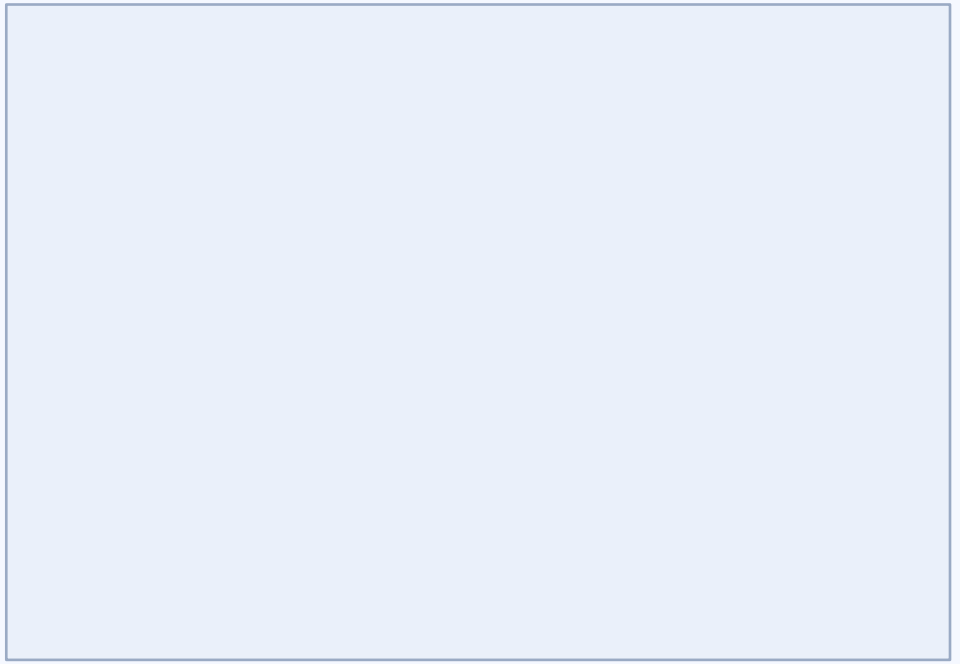
SYMPTOM	CHECK FIRST	EXPECTED EVIDENCE
[What you observe]	[A specific probe or test]	[What a healthy result shows]
[What you observe]	[A specific probe or test]	[What a healthy result shows]
[What you observe]	[A specific probe or test]	[What a healthy result shows]

[What happens next?]

[Ask one question that exposes the mental model.]

[A] [First possibility]

[B] [Second possibility]



[Explain your reasoning.]

[Three things worth remembering]

01 [A short statement learners can use]

02 [A short statement learners can use]

03 [A short statement learners can use]

NEXT / [A small experiment to try independently]

[Datasheets, code and things to read]

[Resource title / author]

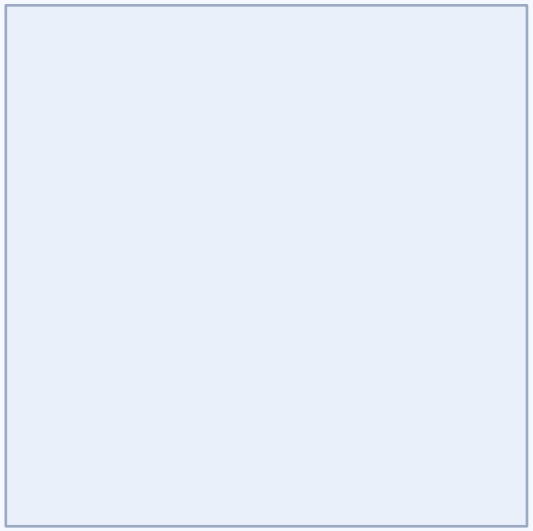
[Exact URL / page / revision]

[Resource title / author]

[Exact URL / page / revision]

[Resource title / author]

[Exact URL / page / revision]



[Short link learners can type]

[Your title]

[A blank page for the next idea.]

OBSERVE

[One short note]

QUESTION

[One short note]

WATCH OUT

[One short note]

→ [label the signal / action]